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HSCAS1.0 API FOR SMS

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2 PREFACE

2.1 Guide

1. This document provides interface standards between HSCAS and SMS and the usage examples for SMS developers.
2. Data transmission between interface of HSCAS and SMS is based on TCP protocol. Default port of HSCAS for SMS connection is 8000. A maximum of 30 SMS clients with different IP addresses is supported.
3. HSCAS only support for synchronous command receiving (When SMS sends a command to HSCAS, it should not send another command until return value received). Asynchronous command is not supported for the present.
4. Command receiving speed of HSCAS is: 50 commands/sec
5. All the parameters are oriented on 32-bit operating system. What need to be clarified is: if SMS is developed on a 64-bit operating system and adopts time type of time_t (8 bytes), the variable should be converted to 4 bytes, and minimum value of time is 1970-01-01 08:00:00 (Note: all date expressions are in the format of yyyy-mm-dd), while maximum is 2038-01-01 00:00:00. All time parameters used as interface parameter in this document are 4 bytes.

2.2 Definitions

Definitions	Meanings
HSCAS	High Security Conditional Access System
SMS	Subscriber Management System

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3 TRANSMISSION DATA FORMAT DEFINITION

HSCAS receives command data packets from SMS, executes relevant process and returns response packet to SMS.

Command data packets and response data packets adopt uniform format. Data packet is transmitted in network order, that is, high-order byte of data is firstly transmitted before low-order byte. Format of data packet is shown as below:

Syntax	Bytes	Notes
SessionID	2	Session number provided by SMS, which is incremental.
CAS_Ver	1	HSCAS version, defined by HSCAS vendor
Command_Type	1	Command type, defined in HSCAS
Data_Len	2	Length of command/response data
Data_Body	Variable (length mentioned above)	Command/response data

Response data packet and its corresponding command data packet differ only in the part of Data_Len and Data_Body.

SessionID: Session number provided by SMS, which is incremental.

CAS_Ver: HSCAS version, defined by CAS vendor; this version is released by Sumavision HSCAS. The current version is 2.

Command_Type: Command type, defined in HSCAS. Corresponding relationship between numerical value and command in HSCAS is as follows:

Command Value	Command Description
0	Reserved
1	Authorizition
2	OSD display
3	Reset PIN
6	Start fingerprint
11	Set terminal status
13	Send Email
13	Start/Stop mandatory message
14	Conditional addressing
15	Set AC value
17	Set area information
18	Cancel urgency broadcasting

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21	Close an account
27	Open an account
31	Clear authorizations of a single terminal
32	Check authorization information of a single terminal
33~255	Reserved

Data_Len: the length of command/response data (in byte).

Data_Body: Data body. Result of data body varies with the type of command. See the chapter of 4 FORMAT DEFINITION OF COMMAND/RESPONSE DATA.

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4 FORMAT DEFINITION OF COMMAND/RESPONSE DATA

1. The following defined data formats are all that of Data_Body in transmission data format definitions.
2. "Expired time" means existing period of the command in HSCAS. After that, HSCAS will stop sending this command. If SMS does display expired time on the interface, it is required to indicate the meaning of expire time to the operator. If not, our suggested time should be adopted.
3. Time interval between opposite operation to same terminal ID (authorization/un-authorization; enable/disable terminal; open/close account) should be longer than 1 second, otherwise HSCAS cannot identify the command clearly. For example, authorization and un-authorization should be separated in time by 1 second at least. For different terminal ID, there is no such limitation.
4. The following "default value" indicates the recommended value when SMS sends parameters to HSCAS.
5. When SMS sends command to a certain terminal ID, and HSCAS returns that terminal account is not opened yet, it is required for SMS to give a prompt box to operator.
6. All terminal ID in the following data format are 4-byte provided by Sumavision. Range of terminal ID is: [0x80000000, 0xFFFFFFFF].

4.1. Open Account

In order to enable a terminal, SMS should send an open card account command for it. The terminal can only receive other commands such like authorization before the terminal receive the open account command successfully.

4.1.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expired time
ChipID_Num	1	1 (only be 1)
ChipID_Len	1	ChipID length (only be 8)
ChipID	8	ChipID value

Description:

Card_ID: terminal ID corresponding to a certain ChipID (**cannot be 0 or 0xFFFFFFFF**).

Expired_Time: expired time of open account command. Recommendation: expire time = send time + 1 day

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ChipID_Num: only be 1

ChipID_Len: only be 8

ChipID: ChipID value. Notice: it should be converted to hexadecimal, and extended with 0 in the header if shorter than 8 byte.

4.1.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Results returned

Result: result of command execution. 0 indicates correct, nonzero is error code.

Error code	Explanation
0xD0000002	SMS sent an incorrect CAS version. When the value of CAS_Ver in packet header does not equal to CAS version, this error code returns.
0xD0000004	Unsupported command type
0xD0000005	Data packet length error
0xD000000A	Terminal ID should not be 0
0xD000001E	Account of this terminal is not opened yet
0xFFFFFFFF	Abnormal communication with CAS (SMS need to re-establish the connection, and send command again)
0xD000000B	Expire time sent by SMS is incorrect
0xD000001B	Terminal ID should not be 0xFFFFFFFF
0xD000001F	ChipID length should only be 8
0xD0000020	Quantity of ChipID should only be 1

4.1.3. Example

4.1.3.1. Send open card account command to a certain terminal

Command purpose:

Send open account command to terminal 0x80000001

Input information:

Current time: 2004-6-1 13:00:00

Expire time: 2004-6-2 13:00:00

Terminal ID: 0x80000001

ChipID: 0x00000A00010A0C00

Command data:

Syntax	Bytes	Notes
Card_ID	4	0x80000001

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Syntax	Bytes	Notes
Expired_Time	4	time_t corresponding to 2004-6-2 13:00:00
ChipID_Num	1	1
ChipID_Len	1	8
ChipID	8	0x00000A00010A0C00

4.2. Close account

In order to close account of a certain terminal, SMS needs to send the command of close account.

4.2.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Is_Send	1	identifier of sending instantly
Expired_Time	4	Expired time

Instructions:

Card_ID: terminal ID corresponding to a certain ChipID **(can not be 0 or 0xFFFFFFFF)**.

Is_Send: identifier of sending instantly, the value is 1.

Expired_Time: expired time of close account command. Recommendation: expired time = send time + 1 day.

4.2.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return result
Emm_Count	1	Quantity of EMM packet returned
for(I=0;I<N;I++)		N=Emm_Count
{		
Emm_Len	1	Length of EMM packet
Emm_Data	Variable (determined by Emm_Len)	Content of EMM packet
}		

Result: result of command execution. 0 indicates correct, nonzero is error code.

Error code	Explanation

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Error code	Explanation
0xD0000002	SMS sent an incorrect HSCAS version. When the value of CAS_Ver in packet header does not equal to HSCAS version, this error code returns.
0xD0000004	Unsupported command type
0xD0000005	Data packet length error
0xD000000A	Terminal ID should not be 0
0xD000001E	Account of this terminal is not opened yet
0xFFFFFFFF	Abnormal communication with CAS (SMS need to re-establish the connection, and send command again)
0xD000000B	Expired time is incorrect
0xD000001B	Terminal ID should not be 0xFFFFFFFF

4.2.3. Example

4.2.3.1. Send cancel card account to a certain terminal

Command purpose:

Send close account to terminal 0x80000001

Input information:

Current time: 2004-6-1 13:00:00

Expire time: 2004-6-2 13:00:00

Terminal ID: 0x80000001

Command data:

Syntax	Bytes	Notes
Card_ID	4	0x80000001
Is_Send	1	1
Expired_Time	4	time_t corresponding to 2004-6-2 13:00:00

4.3. Authorization/un-authorization (delete authorization)

Introduction:

This command is for sending authorization/un-authorization of products to the terminal. When used as un-authorization, the following points should be noticed.

1. With same data structure, difference between authorization and un-authorzation is:

for authorization, Product_Begin_Time < Product_End_Time,

for un-authorization, Product_Begin_Time >= Product_End_Time

2. Authorization adopts end time as identifier.

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3. For authorization or un-authorization, its end time should be later than current system time. If SMS sends a command with the end time earlier than current time, HSCAS will not process and no error will be returned.
4. For a certain terminal ID, authorization and un-authorization of one product ID should not be sent in the data structure of the same command.
5. For a certain terminal ID, expression form of authorization and un-authorization of same product ID is overwrite.

4.3.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
ProductAmount	1	Amount of authorized products
for(i=0;i<N;i++)		N= ProductAmount
{		
Send_Or_Not	1	Send or not, 1 as default
TapingCtrl	1	PVR available or not, 0 as default
ProductNumber	2	Product ID
Product_Begin_Time	4	Start time of authorization
Product_End_Time	4	End time of authorization
Desc_Len	1	Length of description
Description	Variable (determined by Desc_Len)	Description
}		

Guide:

Card_ID: terminal ID corresponding to a certain ChipID (should not be 0 or 0xFFFFFFFF).

ProductAmount: quantity of authorized products (<=10), indicates that this command will send authorization of more than 1 product once.

Send_Or_Not: send or not. Indicates whether to send this authorization to scrambler or not. Default value is 1.

ProductNumber: Product ID. One product may be combined of several services; multiple products may be sent to a certain terminal at one time. The product is defined by SMS.

TapingCtrl: Whether PVR is available or not. 0: not available; 1: available. Default value is 0.

Product_Begin_Time: Start time of authorization. Start time of authorization of the product. Time value means the number of seconds starting from Greenwich time 1970-01-01 00:00:00 (or Beijing time

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1970-01-01 08:00:00). Time value hereinafter refers to this meaning unless otherwise noted.

Product_End_Time: End time of authorization. End time of authorization should not later than current system time.

Desc_Len: length description. Indicates data value (<=255) described below. Default value is 0.

Description: descriptions. EMM packet from higher level CAS should be embedded in, when packaging the authorization EMM packet.

4.3.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Execution result of command
ReturnAmount	1	Quantity of authorization data returned
for(i=0;i<N;i++)		N= ReturnAmount
{		
ProductNumber	2	Product ID
Desc_Len	1	Description length of the data hereinafter
Description	Variable (determined by Desc_Len)	Description
}		

Guide:

Result： Execution result of command. 0: correct. Non-zero: error number.

Error code	Notes
0xD0000002	CAS version is incorrect. When the value of CAS_Ver in the data packet header is not the same with that of CAS, this error returned.
0xD0000004	Unsupported command type.
0xD0000005	Length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Abnormal communication between SMS and CAS (SMS should re-establish the communication and send command again.)
0xD000001B	Terminal ID should not be 0xFFFFFFFF

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Error code	Notes
0xD0000022	Product ID should not be 0

ReturnAmount: quantity of authorization data returned. Default value is 0, only could be nonzero when 2-level CAS exists (only when result is success could this value be bigger than 0; otherwise it will be 0)

ProductNumber: product ID returned

Desc_Len: length of the description data hereinafter

Description: descriptions. HSCAS returns authorization EMM packet to SMS, which will be sent by SMS to lower-CAS for packeting.

4.3.3. Example

4.3.3.1. Sending Authorization to one subscriber

Command Purpose:

Sending authorization of 2 products to the subscriber with terminal 0x80000001:

Authorization of Product 0x101: start time: 2004-1-15 00:00:00; end time: 2004-4-15 23:59:59.

Authorization of Product 0x102: start time: 2003-1-15 00:00:00; end time: 2004-1-15 23:59:59.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
ProductAmount	1	2
Send_Or_Not	1	1
TapingCtrl	1	0
ProductNumber	2	0x0101
Product_Begin_Time	4	time_t corresponding to 2004-1-15 00:00:00
Product_End_Time	4	time_t corresponding to 2004-4-15 23:59:59
Desc_Len	1	0
Send_Or_Not	1	1
TapingCtrl	1	0
ProductNumber	2	0x0102
Product_Begin_Time	4	time_t corresponding to 2003-1-15 00:00:00

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Syntax	Bytes	Value
Product_End_Time	4	time_t corresponding to 2004-1-15 23:59:59
Desc_Len	1	0

4.3.3.2. Sending anti-authorization to one subscriber

Command Purpose:

Sending anti-authorization of the above 2 products to the subscriber with terminal 0x80000001.

Authorization time of the 2 products:

Product 0x101: from 2004-1-15 00:00:00 to 2004-4-15 23:59:59

Product 0x102: from 2003-1-15 00:00:00 to 2004-1-15 23:59:59

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
ProductAmount	1	2
Send_Or_Not	1	1
TapingCtrl	1	0
ProductNumber	2	0x0101
Product_Begin_Time	4	time_t corresponding to 2004-4-15 23:59:59
Product_End_Time	4	time_t corresponding to 2004-4-15 23:59:59
Desc_Len	1	0
Send_Or_Not	1	1
TapingCtrl	1	0
ProductNumber	2	0x0102
Product_Begin_Time	4	time_t corresponding to 2004-1-15 23:59:59
Product_End_Time	4	time_t corresponding to 2004-1-15

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Syntax	Bytes	Value
		23:59:59
Desc_Len	1	0

4.4. Clear authorization of terminal

This command is used by SMS for clear all authorizations of a certain terminal.

4.4.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Send_Or_Not	1	Send or not

Introductions:

Card_ID: terminal ID corresponding to a certain ChipID (**should not be 0 or 0xFFFFFFFF**)

Send_Or_Not: send or not. Indicates if this command will be transferred to scrambler and sent out by scrambler. If yes, this value is 1.

4.4.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return value

Introductions:

Result: result of command execution. 0 indicates success, nonzero is error code.

Error code	Notes
0xD0000002	CAS version is incorrect. When the value of CAS_Ver in the data packet header is not the same with that of CAS, this error returned.
0xD0000004	Unsupported command type.
0xD0000005	Length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Abnormal communication between SMS and CAS (SMS should re-establish the communication and send command again.)
0xD000001B	Terminal ID should not be 0xFFFFFFFF

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4.4.3. Examples

Purpose:

Clear all authorizations on terminal 0x80000001

Command data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Send_Or_Not	1	01

4.5. Query authorization of a certain terminal

This command is used by SMS for acquiring all authorization information of a certain terminal.

4.5.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID

Introductions:

Card_ID: terminal ID corresponding to a certain ChipID **(should not be 0 or 0xFFFFFFFF)**

4.5.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Command execution result
ReturnAmount	2	Quantity of authorization returned
for(i=0;i<N;i++)		N= ReturnAmount
{		
ProductNumber	2	Product ID
BeginTime	4	Start time of authorization
EndTime	4	End time of authorization
}		

Introductions:

Result: result of command execution. 0 indicates success, nonzero is error code.

Error code	Notes

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Error code	Notes
0xD0000002	CAS version is incorrect. When the value of CAS_Ver in the data packet header is not the same with that of CAS, this error returned.
0xD0000004	Unsupported command type.
0xD0000005	Length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Abnormal communication between SMS and CAS (SMS should re-establish the communication and send command again.)
0xD000001B	Terminal ID should not be 0xFFFFFFFF

ReturnAmount: quantity of authorization queried

ProductNumber: product ID

BeginTime: start time of authorization of this product.

EndTime: end time of authorization. End time should be later than current time.

4.5.3. Examples

Query all authorization information of terminal 0x80000001.

Command data:

Syntax	Bytes	Value
Card_ID	4	0x80000001

4.6. OSD display

This command is known as On-Screen-Display, usually used to sending bulletins, renewal reminder, etc.

4.6.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Show_Time_Len	2	Length of display duration
Show_Times	1	Number of times of display
Priority	1	Priority
Expired_Time	4	Expired time

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Syntax	Bytes	Notes
Data_Len	1	Data length
Data _Body	Variable (determined by Desc_Len)	Data

Guide:

Card_ID: terminal ID corresponding to a certain ChipID **(should not be 0)**

Show_Time_Len: Length of display duration (in seconds).The default value is 0. If it is 0, STB will display according to the number of times of display.

Show_Times: Display times. Only when the Show_Time_Len is 0, will this parameter become valid.

Priority: Priority. There are 4 levels in total: low (numeric value is 0); medium (numeric value is 1); high (numeric value is 2) and the highest (numeric value is 3). If subscriber is watching the lower level OSD, when the higher level OSD arrives, the display of lower level OSD will be paused. When the higher level OSD finish the display, shall the lower level OSD continue to be displayed.

Expired_Time: Expired time of OSD display, which means that HSCAS will stop sending the OSD command once the expired time is reached.

Data_Len: Length of data. Length of data hereinafter (larger than 0 and should not exceed 200 byte)

Data_Body: Data. Display content.

Recommended SMS parameter: Show_Time_Len=0; Show_Times=1; Expired_Time = send time + 1 hour;

4.6.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	Length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0
0xD000001E	Account of this terminal is not opened yet.

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Error code	Notes
0xFFFFFFFF	Communication between SMS and CAS is abnormal (SMS should re-establish communication and send command again)
0xD0000003	Length of OSD content is incorrect. When Data_Len in OSD display packet exceed 200, or length is 0, this error returned.
0xD000000B	Expired time error
0xD000000D	OSD priority error
0xD000001D	Show_Time_Len and Show_Times value should not be both 0

4.6.3. Example

Guide:

This command is for notifying terminal to display OSD content.

OSD expired time: the time when CAS stop sending this OSD data. For example: expired time of OSD is 2004-6-1 13:00:00, after 2004-6-1 13:00:00, CAS will not send this OSD content.

4.6.3.1. Enable a terminal to display OSD for certain times

Current time is: 2004-6-1 12:00:00.

Send the subscriber with terminal 0x80000001 and the OSD content is “80000001”; display for 5 times, priority is 0 and expired time is 2004-6-1 13:00:00.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Show_Time_Len	2	0
Show_Times	1	5
Priority	1	0
Expired_Time	4	time_t corresponding to 2004-6-1 13:00:00
Data_Len	1	08
Data _Body	Variable (determined by Desc_Len)	“80000001”

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4.7. Reset PIN

Guide:

This command is used for resetting PIN of a certain terminal. When PIN code is incorrectly input on STB for 3 times, it will be locked. Then this command can be used to reset PIN code of a terminal to default value (currently default value is “12345678”).

4.7.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expired time

Guide:

Card_ID: terminal ID corresponding to a certain ChipID **(should not be 0 or 0xFFFFFFFF).**

Expired_Time: Expired time of PIN reset, that means, CAS will stop sending reset PIN command once the time is reached (suggestion: Expired_Time=send time + 1 day).

4.7.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Nonzero is the error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, this error returns.
0xD0000004	Command type is not supported.
0xD0000005	Incorrect data packet length
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD000000B	Expired time error
0xD000001B	Terminal ID should not be 0xFFFFFFFF

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4.7.3. Example

4.7.3.1. Reset PIN of a certain terminal

Purpose:

Current time is: 2004-6-1 12:00:00

Reset PIN of terminal 0x80000001.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2004-7-1 12:00:00

4.8. Start Fingerprint

SMS sends this command to force terminal ID display on TV.

4.8.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Duration	2	Length of time
Expired_Time	4	Expired time

Guide:

Card_ID: terminal ID corresponding to a certain ChipID (**nonzero**).

Duration: Length of time. Time length of the fingerprint display duration. Unit: second.

Expired_Time: expired time of fingerprint display. CAS will stop sending the command of starting fingerprint once the expired time is reached. (Suggestion: Expired_Time=send time + 1 day).

4.8.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Non-zero is error number.

Error code	Notes

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Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD000000B	Expired time error

4.8.3. Example

4.8.3.1. Start Fingerprint of a certain terminal

For example:

Current time is: 2004-6-1 12:00:00.

Input information:

Send fingerprint to terminal 0x80000001, and duration is 3600 seconds (1 hour); expired time of fingerprint display is 2004-6-2 12:00:00.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Duration	2	3600
Expired_Time	4	Expired time: time_t corresponding to 2004-6-2 12:00:00

4.8.3.2. Start Fingerprint of all terminals

For example:

Current time is: 2004-6-1 12:00:00.

Input information:

Send fingerprint to all subscribers, duration is 3600 seconds (1 hour); expired time of fingerprint display is 2004-6-2 12:00:00.

Command Data:

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Syntax	Bytes	Value
Card_ID	4	0xffffffff
Duration	2	3600
Expired_Time	4	Expired time: time_t corresponding to 2004-6-2 12:00:00

4.9. Set terminal status

This command is normally used by SMS, to set a certain terminal as activated or not activated status.

4.9.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Send_Or_Not	1	Send or not
Card_Status	1	Status type: 0 is not activated, 1 is activated
Expired_Time	4	Expired time

Guide:

Card_ID: terminal ID corresponding to a certain ChipID (can not be 0 or 0xFFFFFFFF).

Send_Or_Not: send or not. Indicates that if this command will be sent to scrambler. This value is 1.

Card_Status: Status type. 0 is not activated, 1 is activated.

Expire_Time: Expire time of start/stop a terminal, that is, CAS will not send this command after this time reached. (Suggestion: Expire_Time=send time +3 days)

4.9.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results
Emm_Count	1	Number of EMM data packet returned
for(I=0;I<N;I++)		N=Emm_Count
{		
Emm_Len	1	Length of EMM data packet
Emm_Data	Variable(dep	Content of EMM

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Syntax	Bytes	Notes
	ends on Emm_Len)	data packet
}		

Result: Execution result of command. Zero is correct. Non-zero is error.

Emm_Count: number of EMM data packet. This value will be 0 if no data returned.

Emm_Len: length of EMM data packet returned.

Emm_Data: content of EMM data packet returned.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Unsupported command type.
0xD0000005	Data packet length error.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD000000B	Expired time error
0xD000000F	Status of terminal error
0xD000001B	Terminal ID should not be 0xFFFFFFFF

4.9.3. Function specifications

4.9.3.1. Set a certain terminal as unactivated status

For example:

Current time is 2004-6-1 12:00:00

Command purpose:

Set terminal 0x80000001 as not activated status, so it will stop working, and expire time is 2004-6-8 12:00:00.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Send_Or_Not	1	01

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Syntax	Bytes	Value
Card_Status	1	0
Expired_Time	4	time_t corresponding to 2004-6-8 12:00:00

4.9.3.2. Set a certain terminal as unactivated status

For example:

Current time is 2004-6-1 12:00:00

Command purpose:

Set terminal 0x80000001 as activated status, so it can work, and expire time is 2004-6-8 12:00:00.

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Send_Or_Not	1	01
Card_Status	1	01
Expired_Time	4	time_t corresponding to 2004-6-8 12:00:00

4.10. Send Email

SMS could send Email to appointed subscribers.

4.10.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expired time
Sender_Name_Len	1	Length of the name of Email sender
Sender_Name	Determined by Sender_Name_Len	Name of Email sender
Email_Content_Len	2	Length of Email content
Email_Content	Determined by Email_Content_Len	Content of Email

Guide:

Card_ID: terminal ID corresponding to a certain ChipID (**non zero**).

Expired_Time: expire time of Email. Suggestion: expired time = send time + 1 day

Sender_Name_Len: length of the name of email sender, with the maximum being 10 bytes and should not be 0.

Sender_Name: Name of Email sender

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Email_Content_Len: length of the content of email, with the maximum being 1000 bytes and should not be 0.

Email_Content: content of Email

4.10.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD0000006	Length of name of Email sender error
0xD0000007	Length of Email content error
0xD000000B	Expired time error

4.10.3. Example

4.10.3.1. Send emails to a certain card

For example:

Command Purpose:

Send Email to card 0x80000001.

Input information:

Current time is: 2004-6-1 13:00:00.

Expired time is 2004-6-2 13:00:00

Terminal ID: 0x80000001

Email sender: DVT

Email content: Email from DVT

Command Data:

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Syntax	Bytes	Value
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2004-6-2 13:00:00
Sender_Name_Len	1	3
Sender_Name	Determined by Sender_Name_Len	DVT
Email_Content_Len	2	0e
Email_Content	Determined by Email_Content_Len	Email from DVT

4.11. Start/Stop mandatory message

SMS could send mandatory command or cancel it to appointed subscribers.

4.11.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expire time
Sender_Name _Len	1	Length of name of mandatory message sender
Sender_Name	Determined by Sender_Name_Len	Name of mandatory message sender
Msg_Content _Len	2	Length of mandatory message content
Msg_Content	Determined by Email_Content_Le n	mandatory message content, keep the content blank to cancel mandatory message
Msg_Flag	1	Identifier of mandatory message, the value is 2.

Note:

Card_ID: corresponding to a certain ChipID (**nonzero**).

Expired_Time: expire time of mandatory message. Suggestion: expire time = send time +1.

Sender_Name_Len: length of name of mandatory message sender, with the maximum being 10 bytes and should not be 0.

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Sender_Name: name of mandatory message sender.

Msg_Content_Len: length of mandatory message content, with the maximum being 1000 bytes.

Msg_Content: content of mandatory message.

Msg_Flag: identifier of mandatory message, the value is 2.

4.11.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD0000006	Length of name of mandatory message sender error
0xD0000007	Length of mandatory message content error
0xD000000B	Expired time error
0xD000000D	Identifier of mandatory message error

4.11.3. Example

4.11.3.1. Send mandatory message to a certain terminal

Example:

Command purpose:

Send mandatory message to terminal 0x80000001

Input information:

Current time: 2010-4-20 13:00:00

Expire time: 2010-4-21 13:00:00

Terminal ID: 0x80000001

Sender of mandatory message: DVT

Content of mandatory message: mandatory message

Identifier of mandatory message: 2

Command data:

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Syntax	Bytes	Value
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2010-4-21 13:00:00
Sender_Name _Len	1	3
Sender_Name	Determined by Sender_Name_Len	DVT
Msg_Content _Len	2	0x11
Msg_Content	Determined by Email_Content_Len	Mandatory message
Msg_Flag	1	2

4.11.3.2. Cancel mandatory message of a certain message

Example:

Command purpose:

Cancel mandatory message of terminal 0x80000001

Input information:

Current time: 2010-4-20 13:00:00

Expire time: 2010-4-21 13:00:00

Terminal ID: 0x80000001

Sender of mandatory message: DVT

Content of mandatory message: <blank>

Identifier of mandatory message: 2

Command data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2010-4-21 13:00:00
Sender_Name _Len	1	3
Sender_Name	Determined by Sender_Name_Len	DVT
Msg_Content _Len	2	0
Msg_Content	Determined by Email_Content_Len	<blank>

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Syntax	Bytes	Value
Msg_Flag	1	2

4.12. Conditional addressing

4.12.1. Command Data Format

Syntax	Bytes	Notes
Condition_Len	2	Length of condition
Condition	Determined by Condition_Len	Content of condition
Condition_Cmd_Type	1	Command type requiring conditional addressing
Condition_Cmd_Content	Different commands have different formats.	Command content requiring conditional addressing

Condition_Len: Length of condition. The maximum length is 160 bytes.

Condition: Content of condition, details as follow:

Now SMS supports addressing such as terminal ID (0x23), region ID (0x24), ChipID (0x53).

8 operational character: <, <=, >, >=, =, !=, &&, || with the following corresponding values:

<(0x13), <=(0x14), >(0x15), >=(0x16), =(0x17), !=(0x18),&&(0x1c),||(0x1d)

Example 1: Send Emails to the terminal with region ID 0x1001

The corresponding condition is '2417040000 1001', among which '00001001' representing hexadecimal region ID value, '04' is the length of region ID.

Example 2: Send Emails to the terminals with the region ID >= 0x1001

The corresponding condition is '23160400001001', among which '00001001' representing hexadecimal region ID value, '04' is the length of region ID.

Example 3: Send Emails to the terminals with the region ID >=0x1001 and terminal IDs <0x1000

The corresponding condition is '241604000010011c23130400001000'.

[Notice] one command should use no more than 5 conditions.

Condition_Cmd_Type: Command type requiring conditional addressing. Relationship between the numeric value of Condition_Cmd_Type and command is as follows:

Numeric value	Corresponding command
1	Conditional Email
2	Conditional OSD
3	Conditional restriction of broadcasting
4	Cancel conditional restriction of broadcasting
5	Urgency broadcasting

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Numeric value	Corresponding command
7~255	Reserved

Condition_Cmd_Content: Command content requiring conditional addressing. The data format is determined by Condition_Cmd_Type.

4.12.1.1. Conditional Email

Syntax	Bytes	Notes
Expired_Time	4	Expired time
Sender_Name_Len	1	Length of the name of Email sender
Sender_Name	Determined by Sender_Name_Len	Name of Email sender
Email_Content_Len	2	Length of Email content
Email_Content	Determined by Email_Content_Len	Content of Email

Guide: (see section 4.12 Send Emails)

Expired_Time: expired time of EMM data packet. Suggestion: expired time = send time + 1day.

Sender_Name_Len: Length of the name of Email sender, with the maximum being 10 bytes and should not be 0.

Sender_Name: Name of Email sender

Email_Content_Len: Length of the content of Email, with the maximum being 1024 bytes and should not be 0.

Email_Content: the content of Email

Response data format of conditional Email

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).

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Error code	Notes
0xD0000006	Length of name of Email sender error
0xD0000007	Length of Email content error
0xD0000008	Condition error. When length or expression form of condition is incorrect, returns this error.
0xD000000B	Expire time error.
0xD000000D	Identifier of Email error.

4.12.1.2. Conditional OSD

Syntax	Bytess	Notes
Expired_Time	4	Expired time
Show_Time_Len	2	Length of display period
Show_Times	1	Display number of times
Priority	1	Priority
Data_Len	1	Length of data
Data _Body	Variable (determined by Desc_Len)	Data

Guide: (see section 4.2 OSD display)

Show_Time_Len: Length of display period, unit: second. The default value is 0. If it is 0, STB will display according to the number of times.

Show_Times: Display number of times. The default value is 3. The maximum value is 256; if it exceeds the value, only valid value is used.

Priority: priority level. There are 4 levels in total: low (numeric value is 0), medium (numeric value is 1), high (numeric value is 2) and the highest (numeric value is 3).

Expired_Time: expired time of OSD display, i.e., CAS will stop sending the OSD command once the time is reached. Suggestion: expired time = send time +1 day.

Data_Len: length of data. Length of data hereinafter (0< length <200).

Data_Body: data. OSD display content.

Response data format of conditional OSD

Syntax	Bytes	Notes
Result	4	Return results

Result: Execution result of command. 0: correct. Non-zero: an error number.

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Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD0000003	Length of OSD content error. When the value of Data_Len of OSD display data packet exceeds 200 or when the length is 0, returns this error.
0xD0000008	Condition error. When length or expression form of condition is incorrect, returns this error.
0xD000000B	Expire time error.
0xD000000D	Priority of OSD error.
0xD000001D	Time length and number of times should not be both 0.

4.12.1.3. Conditional restriction of broadcasting

Syntax	Bytes	Notes
ProductNumber	2	Product ID
Expired_Time	4	Expired time

Guide:

ProductNumber: product id. A product may be combined of several services. The product is defined by SMS.

Expired_Time: expired time of conditional restriction of broadcasting, i.e., CAS will stop sending this conditional restriction of broadcasting command once the time is reached.

Response data format of conditional restriction of broadcasting

Syntax	Bytes	Notes
Result	4	Return result

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes

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Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD0000008	Condition error. When length or expression form of condition is incorrect, returns this error.
0xD000000B	Expire time error.

4.12.1.4. Cancel conditional restriction of broadcasting

Syntax	Bytes	Notes
ProductNumber	2	Product ID

Guide:

ProductNumber: product ID. A product may be combined of several services. The product is defined by SMS.

Response data format of conditional restriction of broadcasting

Syntax	Bytes	Notes
Result	4	Return result

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).

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Error code	Notes
0xD0000008	Condition error. When length or expression form of condition is incorrect, returns this error.

4.12.1.5. Conditional urgency broadcasting

Syntax	Bytes	Notes
ID	4	ID of urgency broadcasting
Expired_Time	4	Expired time
Ori_Net_ID	2	Original network ID
TS_ID	2	Transport Stream ID
Service_ID	2	Service ID
Show_Time_Len	2	Duration
Msg_Len	1	Length of urgency message
Msg_Content	Variable (determined by Msg_Len)	Content of urgency message

Guide:

ID: ID of urgency broadcasting; if the ID of two broadcastings are identical, the latter one will overwrite the former one.

Expired_Time: expired time of urgency broadcasting, i.e., CAS will stop sending the "Conditional urgency broadcasting" command once the time is reached. The expired time should not be less than the current time. End time of urgency broadcasting should be set according to customer's requirement.

Ori_Net_ID: Original network ID.

TS_ID: transport stream ID.

Service_ID: Service ID.

Show_Time_Len: Display time duration of the message.

Msg_Len: Length of urgency message, with the maximum not exceeding 100 bytes

Msg_Content: Content of urgency message.

4.12.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes

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Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD0000008	Conditional address error. When length of condition or expression form error, returns this error.
0xD000000B	Expired time error
0xD0000011	Content of urgency broadcasting should not exceed 100 byte.
0xD000001D	Time duration should not be 0

4.12.3. Example

4.12.3.1. Send conditional Email

For example:

Command Purpose:

Send emails to the terminals with region ID of 0x1001.

Command Data:

Syntax	Bytes	Value (Hexadecimal)
Condition_Len	2	00 07
Condition	Determined by Condition_Len	Conditional expression of region ID=0x1001: 24 17 04 00 00 10 01
Condition_Cmd_Type	1	01
Expired_Time	4	time_t corresponding to 2004-7-1 13:00:00
Sender_Name_Len	1	03
Sender_Name	Determined by Sender_Name_Len	DVT
Email_Content_Len	2	00 0e
Email_Content	Determined by Email_Content_Len	Email from DVT

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4.12.3.2. Send conditional OSD display

For example:

Command Purpose:

Send OSD to the terminals with terminal ID smaller than 0x10000000

Command Data:

Syntax	Bytes	Value (Hexadecimal)
Condition_Len	2	00 07
Condition	Determined by Condition_Len	Conditional expression for terminal ID < 0x10000000: 23 13 04 10 00 00 00
Condition_Cmd_Type	1	2
Expired_Time	4	time_t corresponding to 2004-7-1 13:00:00
Show_Time_Len	2	00 40
Show_Times	1	1
Priority	1	0
Data_Len	1	3
Data _Body	Variable (determined by Desc_Len)	DVT

4.12.3.3. Send conditional restriction of broadcasting

For example:

Command Purpose:

Send the conditional restriction of broadcasting to the terminals with terminal ID smaller than 0x10000000.

Command Data:

Syntax	Bytes	Value (Hexadecimal)
Condition_Len	2	00 07
Condition	Determined by Condition_Len	Conditional expression for terminal ID < 0x10000000: 23 13 04 10 00 00 00
Condition_Cmd_Type	1	0x3
ProductNumber	2	00 03

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Syntax	Bytes	Value (Hexadecimal)
Expired_Time	4	time_t corresponding to 2004-7-1 13:00:00

4.12.3.4. Cancel conditional restriction of broadcasting

For example:

Command Purpose:

Cancel conditional restriction of service No.3 broadcasting of terminal 0x80000001.

Command Data:

Syntax	Bytes	Value
Condition_Len	2	00 07
Condition	Determined by Condition_Len	Conditional expression for terminal = 0x80000001: 3B 17 04 80 00 00 01
Condition_Cmd_Type	1	0x04
ProductNumber	2	00 03

4.12.3.5. Send conditional urgency broadcasting

For example:

Command Purpose:

Send urgency broadcasting to all terminals

Command Data:

Syntax	Bytes	Value
Condition_Len	2	7
Condition	Determined by Condition_Len	Addressing condition of 'Terminal ID > = 1': 23 16 04 00 00 00 01
Condition_Cmd_Type	1	0x05
ID	4	00 00 00 01
Expired_Time	4	time_t corresponding to 2004-7-1 13:00:00
Ori_Net_ID	2	00 01
TS_ID	2	00 01
Service_ID	2	00 01
Show_Time_Len	2	00 30
Msg_Len	1	3
Msg_Content	Variable	DVT

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Syntax	Bytes	Value
	(determined by Msg_Len)	

4.13. Set AC Information (Set user-defined information)

SMS uses this command to send the operator-defined information to the terminals for special purposes. For example, when some terminals are used as VIP, STB can read the VIP identifier, then find corresponding BAT table according to the identifier, where acceptable digital TV programs will be described respectively. In this way, the following purposes can be achieved: with VIP terminals, subscribers can watch all domestic and overseas programs, while ordinary terminals can only watch ordinary programs. You can also set the subscriber digital TV certificate No, subscriber level and industry category, etc.

4.13.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expired time
ACData	100	AC Data

Guide:

Card_ID: terminal ID corresponding to a certain ChipID. (should not be 0 or 0xFFFFFFFF).

Expired_Time: Expire time of AC information, that is, CAS will not send this command of Setting AC Information any more after the expiration time (Recommended Expired_Time=Sending time + 7 days).

ACData: AC data. Since ACData must be 100 bytes, the remaining should be supplemented by 0. The valid AC data has 5 bytes, with the first byte standing for data type (0xC8~0xFE) and the following 4 bytes referring to the self-defined data value.

4.13.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes

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Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD000000A	Card number should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD000000B	Expired time error

4.13.3. Functions specifications

4.13.3.1. Set self-defined value of a certain terminal separately

For example:

Command purpose:

Set self-defined value (0x00001001) of the terminal 0x80000001.

Input information:

Current time: 2004-6-1 13:00:00

Expiration time: 2004-6-8 13:00:00

Terminal ID: 0x80000001

AC data: C8 00 00 10 01

Command data:

Syntax	Bytes	Notes
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2004-6-8 13:00:00
ACData	100	C8 00 00 10 01 00 00 00 00 00 00 00 00 00 00 …… 00 00 00

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4.14. Set region information

Fucntion: modify region ID of the terminal. When operator use regional lock function, region ID of terminals should be changed to actual local region ID.

4.14.1. Command Data Format

Syntax	Bytes	Notes
Card_ID	4	Terminal ID
Expired_Time	4	Expired time
Area	4	Region ID

Guide:

Card_ID: terminal ID corresponding to a certain ChipID **(should not be 0 or 0xFFFFFFFF)**.

Expired_Time: Set expired time of region ID information, i.e., CAS will stop sending the "Set region ID" command once the time is reached (Suggestion: Expired_Time = send time + 7 days).

Area: region ID.

4.14.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD000000A	Terminal ID should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).
0xD000000B	Expired time error
0xD000001B	Terminal ID should not be 0xFFFFFFFF

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4.14.3. Example

4.14.3.1. Set region information for a certain terminal

For example:

Command Purpose:

Set region information of terminal 0x80000001 as 0x1001

Input information:

Current time is: 2004-6-1 13:00:00.

Expired time is 2004-6-4 13:00:00

Terminal ID: 0x80000001

Region ID: 0x1001

Command Data:

Syntax	Bytes	Value
Card_ID	4	0x80000001
Expired_Time	4	time_t corresponding to 2004-6-4 13:00:00
Area	4	0x1001

4.15. Cancel urgency broadcasting

This command is the cancel command of conditional urgency broadcasting.

4.15.1. Command Data Format

Syntax	Bytes	Notes
ID	4	ID of urgency broadcasting

4.15.2. Response Data Format

Syntax	Bytes	Notes
Result	4	Return results

Guide:

Result: Execution result of command. 0: correct. Non-zero: an error number.

Error code	Notes
0xD0000002	The CAS version is wrong. When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.

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Error code	Notes
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).

4.15.3. Example

4.15.3.1. Cancel urgency broadcasting

For example:

Command Purpose:

Cancel the urgency broadcasting in the previous example.

Command Data:

Syntax	Bytes	Value
ID	4	1

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5 ANNEX A:ERROR NUMBER DEFINITION

Error Number	Interpretations
0xD0000002	The CAS version is wrong.When the value of CAS_Ver in the data packet header is not the same with that of CAS, returns this error.
0xD0000003	The length of OSD is wrong. When the value of Data_Len of OSD display data packet exceeds 200 or when the length is 0, returns this error.
0xD0000004	Command type is not supported.
0xD0000005	The length of the data packet is wrong.
0xD0000006	The length of the Email sender name is wrong.
0xD0000007	The length of the Email content is wrong.
0xD0000008	The condition is wrong. When the length of the condition or the format of expression is wrong, returns this error.
0xD000000A	Terminal ID should not be 0.
0xD000000B	The expired time is wrong.
0xD000000D	The priority is wrong.
0xD000000F	The status of terminal (activation/de-activation) is wrong.
0xD0000011	The content of urgency broadcasting should not exceed 100 bytes.
0xD000001B	Terminal ID should not be 0xFFFFFFFF.
0xD000001D	Invalid data. The Show_Time_Len and Show_Times value should not be both 0; time length of conditional urgency broadcasting should not be 0.
0xD000001E	Account of this terminal is not opened yet.
0xD000001F	ChipID's length should only be 8.
0xD0000020	Quantity of ChipID should only be 1.
0xD0000022	Product ID should not be 0.
0xFFFFFFFF	Communication between SMS and CAS is abnormal. (SMS should re-establish communication, and send command again).

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6 VERSION CONTROL

Version	Editer	Edit time	Description
V1.0.0	Liu cheng	2017/09/26	Create

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